

Alfido Tech Sales Performance Analysis Report

Project Title

Sales Performance Analysis Using Python and Power BI

1. Introduction

The purpose of this project is to analyze sales performance across different regions, product categories, customer segments, and time periods using sales data. The project focuses on identifying business trends, top-performing products, weak-performing products, and seasonal sales patterns.

The analysis was performed using Python for data cleaning and exploratory analysis, and Power BI for creating an interactive dashboard.

2. Project Objectives

The main objectives of this project are:

- Clean and prepare the sales dataset
 - Analyze revenue performance across regions and categories
 - Identify top-performing and low-performing products
 - Study monthly and yearly sales trends
 - Understand customer segment performance
 - Create KPI metrics and visual dashboards
 - Suggest business improvements for increasing sales
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3. Tools and Technologies Used

Tool	Purpose
Python	Data cleaning and analysis
Pandas	Data manipulation
Matplotlib & Seaborn	Data visualization
Power BI	Interactive dashboard creation
Jupyter Notebook	Analysis notebook

4. Dataset Description

The dataset used for this project contains sales transaction information.

Dataset Columns

- Row_ID

- Order_ID
- Order_Date
- Ship_Date
- Ship_Mode
- Customer_ID
- Customer_Name
- Segment
- Country
- City
- State
- Postal_Code
- Region
- Product_ID
- Category
- Sub_Category
- Product_Name
- Sales
- Year
- Month
- Month_Name

5. Data Cleaning and Preparation

The following preprocessing steps were performed:

- Removed duplicate records
- Checked and handled missing values
- Converted Order_Date into proper datetime format
- Created additional date columns such as Year, Month, and Month_Name
- Verified column names and data types

The dataset was cleaned successfully and prepared for analysis.

6. KPI Metrics Created

The following Key Performance Indicators (KPIs) were created:

KPI	Description
Total Revenue	Total sales generated
KPI	Description
Total Orders	Number of unique orders
Average Sales	Average sales value
Total Customers	Number of unique customers
Average Order Value	Revenue generated per order

7. Exploratory Data Analysis

7.1 Monthly Sales Trend

A line chart was created to analyze monthly sales performance and identify seasonal trends.

Observations

- Sales fluctuated across months.
 - Some months showed significantly higher sales indicating seasonal demand.
 - Peak sales periods can help improve inventory planning.
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7.2 Regional Sales Performance

A regional sales analysis was performed using bar charts.

Observations

- Some regions generated much higher sales compared to others.
 - High-performing regions contribute significantly to total revenue.
 - Low-performing regions require additional marketing focus.
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7.3 Category Analysis

Sales performance was analyzed across product categories.

Observations

- Certain categories contributed more revenue.
 - Product category performance can guide business investment decisions.
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7.4 Top Products Analysis

Top-selling products were identified using sales-based ranking.

Observations

- A small number of products contributed a major portion of revenue.
 - Best-selling products should receive higher promotional focus.
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7.5 Customer Segment Analysis

Sales were analyzed based on customer segments.

Observations

- Consumer segment generated the highest sales.
 - Segment analysis helps understand customer purchasing behavior.
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7.6 Shipping Mode Analysis

Shipping modes were analyzed to understand delivery preferences.

Observations

- Standard shipping mode was used most frequently.
 - Delivery preferences influence operational planning.
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8. Dashboard Development

An interactive dashboard was developed in Power BI.

Dashboard Features

- KPI Cards
- Monthly Sales Trend
- Regional Performance Analysis
- Category Performance Charts
- Top Product Visualization
- Customer Segment Analysis
- Matrix Visualization
- Interactive Filters and Slicers

The dashboard allows users to interactively explore sales performance.

9. Business Insights

The analysis generated several important business insights:

- Certain regions consistently generated higher revenue.

- A few products contributed significantly to overall sales.
 - Seasonal sales peaks were observed during specific months.
 - Customer purchasing patterns varied across segments.
 - Some product categories showed lower sales performance.
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10. Tactical Improvements for Alfido Tech

1. Increase Marketing in High-Performing Regions

Invest more marketing resources in regions with strong sales performance to maximize revenue growth.

2. Promote Best-Selling Products Aggressively

Use advertisements, discounts, and bundles to further increase sales of top-performing products.

3. Improve Low-Performing Products and Categories

Analyze customer preferences and improve pricing or promotions for weak-performing products.

4. Prepare Inventory Before Peak Seasons

Maintain sufficient inventory before high-demand months to avoid stock shortages.

5. Introduce Customer Loyalty Programs

Provide reward programs and personalized offers to increase customer retention and repeat purchases.

11. Challenges Faced

- The dataset did not contain profit-related columns.
 - Conversion metrics could not be directly calculated.
 - Date formatting issues required preprocessing.
 - Some visualizations required sorting adjustments.
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12. Conclusion

This project successfully analyzed sales performance using Python and Power BI. The analysis helped identify sales trends, top-performing products, regional performance, and customer behavior patterns.

The interactive dashboard and insights can support business decision-making and help improve sales strategies for Alfido Tech.

The project demonstrates practical skills in:

- Data Cleaning
- Exploratory Data Analysis
- Business Intelligence

- Dashboard Development
 - Data Visualization
 - Business Insight Generation
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13. Future Improvements

Future enhancements can include:

- Adding profit and cost analysis
 - Including customer retention metrics
 - Predictive sales forecasting
 - Real-time dashboard integration
 - Machine learning-based sales prediction
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14. Deliverables

The following deliverables were prepared:

- Jupyter Notebook with analysis and charts
 - Power BI Dashboard
 - PDF Summary Report
 - Cleaned Sales Dataset
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15. References of Dataset

<https://www.kaggle.com/datasets/bhanupratapbiswas/superstore-sales>
